Special types of concrete made with portland cement		
Architectural concrete	High-early-strength concrete	Roller-compacted concrete
Autoclaved cellular concrete	High-performance concrete	Sawdust concrete
Centrifugally cast concrete	High-strength concrete	Self-compacting concrete
Colloidal concrete	Insulating concrete	Shielding concrete
Colored concrete	Latex-modified concrete	Shotcrete
Controlled-density fill	Low-density concrete	Shrinkage-compensating concrete
Cyclopean (rubble) concrete	Mass concrete	Silica-fume concrete
Dry-packed concrete	Moderate-strength lightweight concrete	Soil-cement
Epoxy-modified concrete	Nailable concrete	Stamped concrete
Exposed-aggregate concrete	No-slump concrete	Structural lightweight concrete
Ferrocement	Pervious (porous) concrete	Superplasticized concrete
Fiber concrete	Photocatalytic Concrete	Terrazzo
Fill concrete	Polymer-modified concrete	Tremie concrete
Flowable fill	Pozzolan concrete	Ultra High Performance Concrete
Flowing concrete	Precast concrete	Vacuum-treated concrete
Fly-ash concrete	Prepacked concrete	Vermiculite concrete
Gap-graded concrete	Preplaced aggregate concrete	White concrete
Geopolymer concrete	Reactive-powder concrete	Zero-slump concrete
Heavyweight concrete	Recycled concrete	
Special types of concrete not using portland cement		
Acrylic concrete	Gypsum concrete	Polymer concrete
Aluminum phosphate concrete	Latex concrete	Potassium silicate concrete
Calcium aluminate concrete	Magnesium phosphate concrete	Sodium silicate concrete
Epoxy concrete	Methyl methacrylate (MMA) concrete	Sulfur concrete
Furan concrete	Polyester concrete	

For more information on special types of concrete, please refer to http://www.concrete.org/Technical/CCT/ACI-Terminology.aspx

Special concrete

Overview

- Concrete is most vital material in modern construction.
- In addition to normal concrete, other varieties in use are, high strength and high performance concrete, self compacting, light weight, high density, fibre reinforced, polymer, coloured concrete etc.
- The making of concrete is an art as well as a science.



Introduction

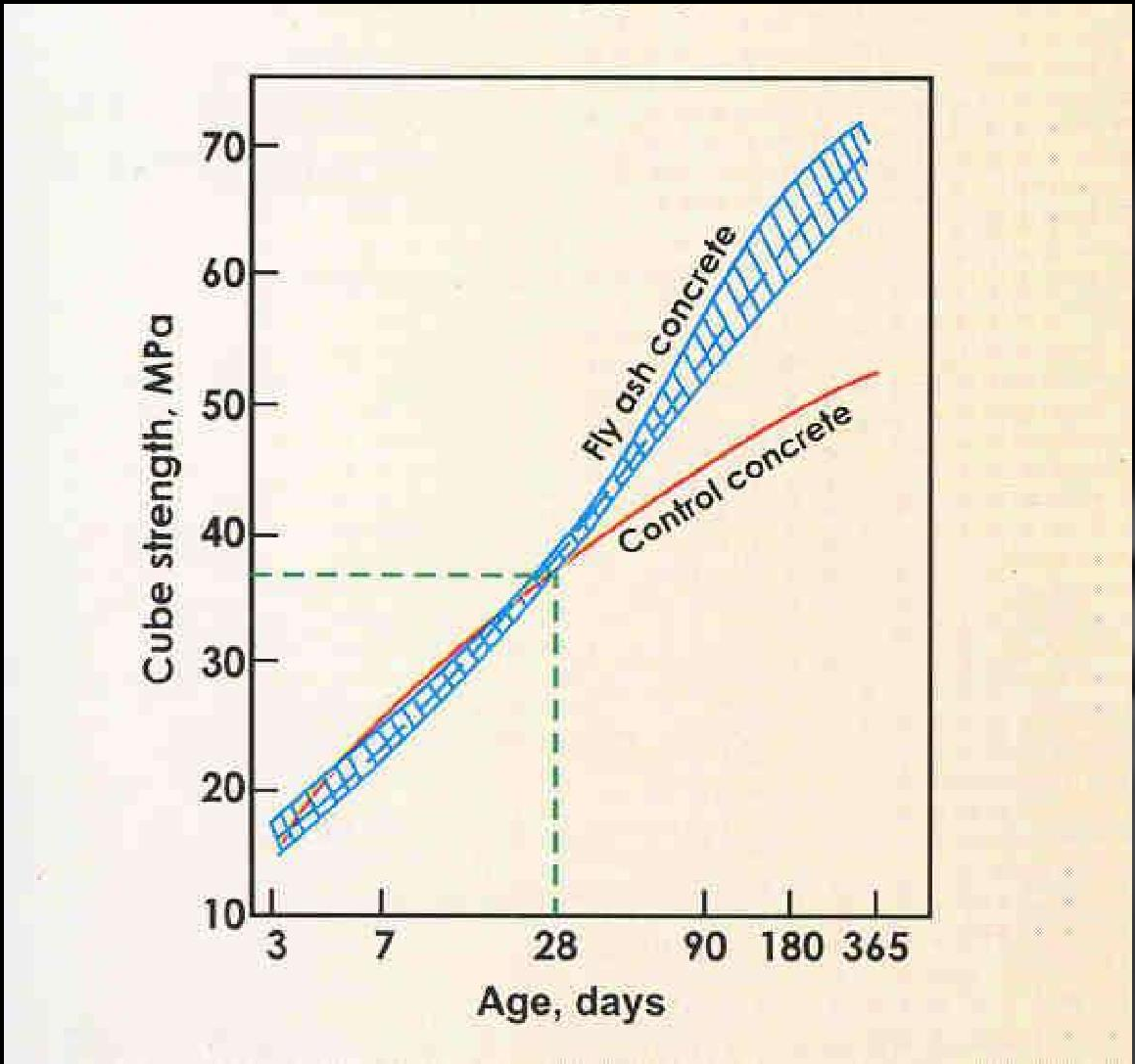
- Special types of concrete are those with out-of-the- ordinary properties or those produced by unusual techniques.
- Concrete is by definition a composite material consisting essentially of a binding medium and aggregate particles, and it can take many forms.
- These concretes do have advantages as well as disadvantages.

Types of special concrete

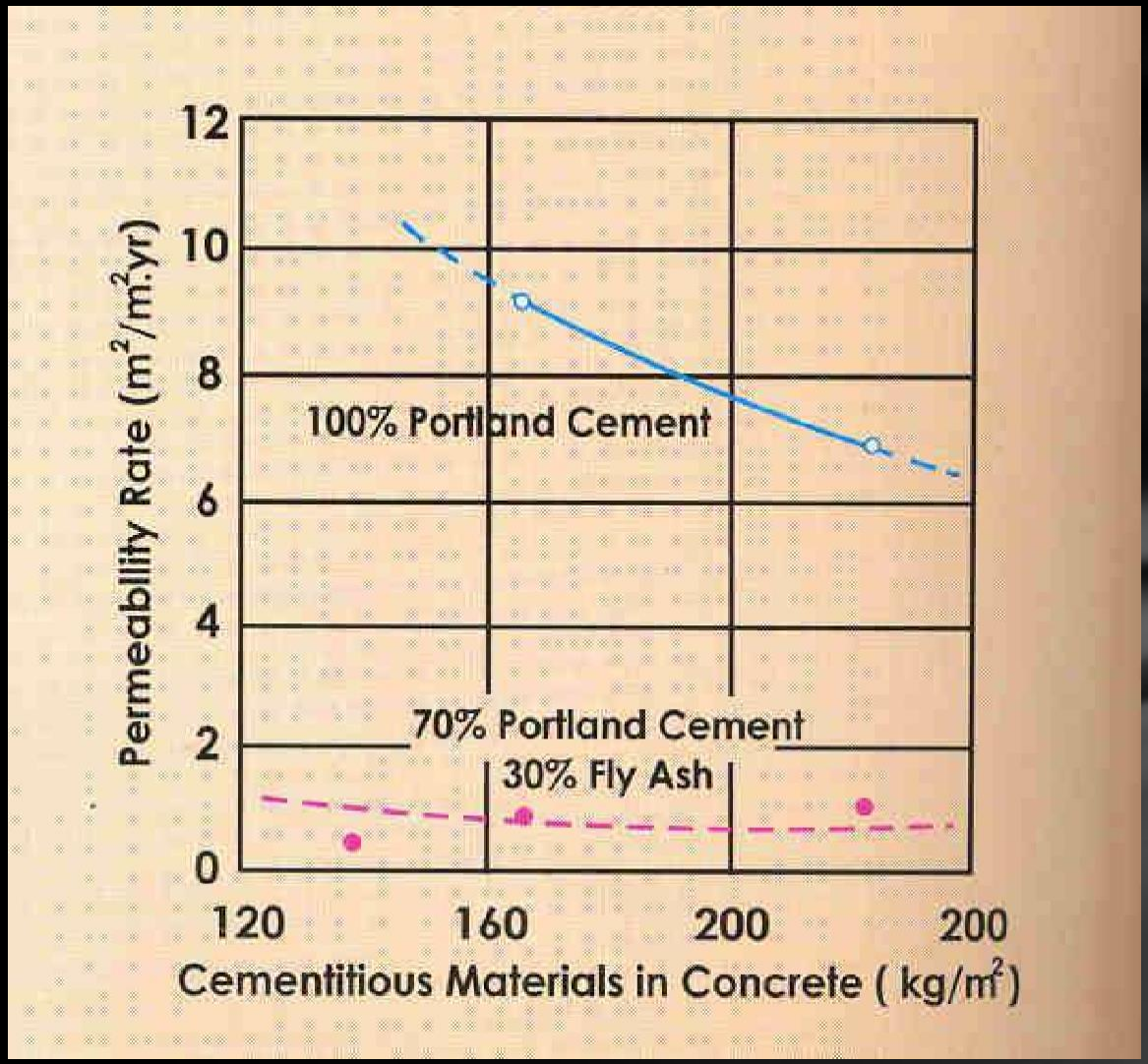
1. High Volume Fly Ash Concrete.
2. Silica fume concrete.
3. GGBS, Slag based concrete.
4. Ternary blend concrete.
5. Light weight concrete.
6. Polymer concrete.
7. Self Compacting Concrete.
8. Coloured Concrete.
9. Fibre-reinforced Concrete.
10. Pervious Concrete.
11. Water-proof Concrete.
12. Temperature Controlled Concrete.

1.High Volume Fly Ash Concrete.

- Is used to replace a portion of the portland cement used in the mix.
- According to IS: 456 – 2000 **replacement of OPC by Fly-ash up to 35% as binding material is permitted.**
- HVFAC is a concrete where excess of 35%of fly-ash is used as replacement.
- Use of fly ash is because of many factors such as
  a) Abundance of fly ash i.e. 110million tons of fly ash is produced in India every year.
  b) Fly ashes from major TPP are of very high quality i.e. quality of fly ash.
  c) Economic factor i.e. Cost of fly ash with in 200 km from a TPP is as low as 10% to 20% of the cost of cement.
  d) Environmental factors i.e. reduction in CO_2 emission.



Strength relation between control concrete and fly ash concrete



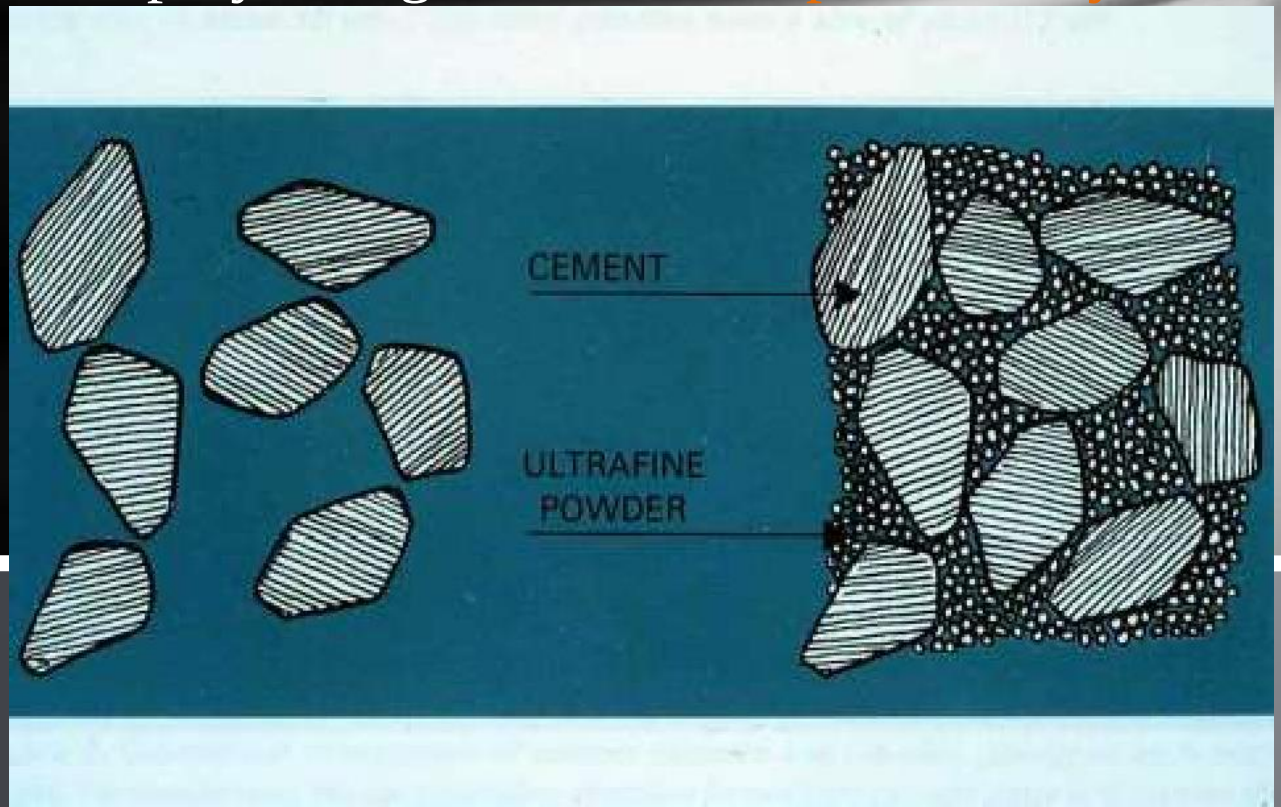
Permeability relation between control and fly ash concrete.

2.Silica fume concrete

- Very fine non-crystalline silica produced in electric arc furnaces as a by product.
- Highly reactive pozzolan used to improve mortar and concrete.
- Silica fume in concrete produces two types of effect viz.
  - Physical effect
  - Chemical effect
- The transition zone is a thin layer between the bulk hydrated cement paste and the aggregate particles in concrete. This zone is the weakest component in concrete, and it is also the most permeable area. Silica fume plays a significant role in the transition zone through both its physical and chemical effects.

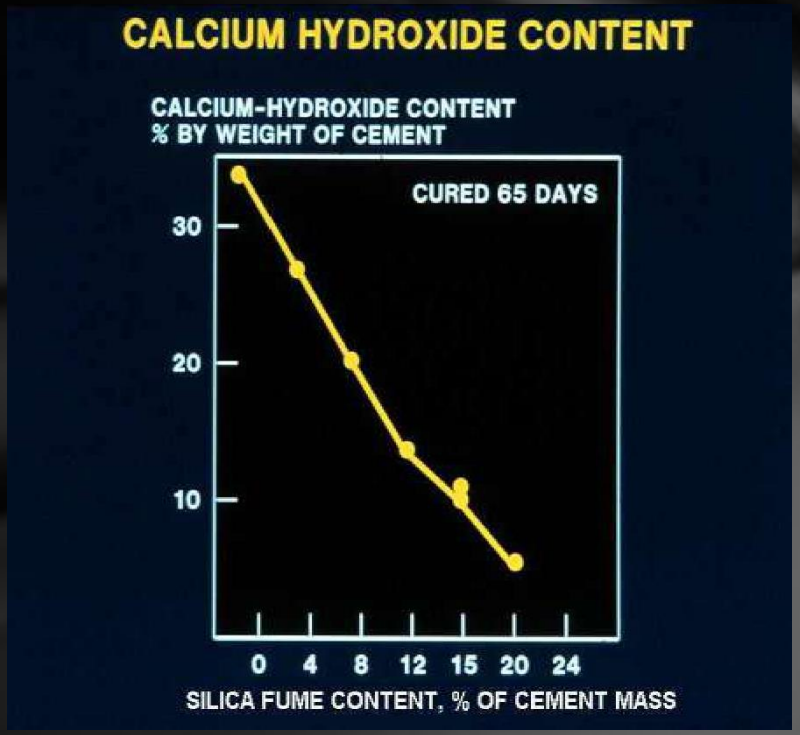
Physical Effect

- The presence of any type of very small particles will improve concrete properties. This effect is termed either "particle packing" or "micro filling".
- physical mechanisms do play a significant role, particularly at early ages.



Chemical Effect

- Silica fume is simply a very effective pozzolanic material.
- pozzolanic means a siliceous or siliceous and aluminous material, which in itself possess little or no cementious value but will, in finely divided form and in the presence of moisture, chemically react with calcium hydroxide at ordinary temperatures to form compounds possessing cementious properties.



3. GGBS, Slag based concrete

- By-product of the iron manufacturing industry, replacement of Portland cement with GGBS will lead to significant reduction of carbon dioxide gas emission.
- GGBS powder is almost white in colour in the dry state Fresh GGBS concrete may show mottled green or bluish-green areas on the surface mainly due to the presence of a small amount of sulphide.
- GGBS concrete requires longer setting times than Portland cement concrete, probably due to the smooth and glassy particle forms of GGBS. If the temperature is 23oC or replacement level of portland cement by GGBS is less than 30% , the setting times will not significantly be affected.
- when GGBS replacement level is less than 40%, bleeding is generally unaffected. At higher replacement levels, bleeding rates may be higher.

- GGBS concrete has lower early strengths because the rate of initial reaction of GGBS is slower than that of Portland cement.
- GGBS is therefore generally grounded to a finer state than Portland cement i.e. from around 4000 cm2/g to 6000 cm2/g resulting in significant increase in 28- day strength.
- It was also reported that the early strengths (up to 28 days) of concrete mixes (with 25%, 35%, 50%, and 60% GGBS replacements) were lower than that of Portland cement concrete mixes. By 56 days, the strength of 50% and 60% GGBS mixes exceeded that of the Portland cement mix, and by one year all GGBS mixes were stronger than the Portland cement mixes.
- Due to its longer setting time, it can be transported to distant places but care should be taken while casting because there are chances that bleeding may take place.



Ground granulated blast furnace slag

4. Ternary blend concrete

- Ternary concrete mixtures include three different cementitious materials i.e. combinations of portland cement, slag cement, and a third cementitious material. The third component is often fly ash, but silica fume is also common.
- Other material in combination with portland and slag cement, such as rice husk ash are not currently in common usage.
- Slag cement has been used in ternary mixtures for decades.

Benefits

1. High strength
2. Low permeability
3. Corrosion resistance
4. Sulphate resistance
5. ASR resistance
6. Elimination of thermal cracking

Ternary mixtures can be used and have been used in virtually any concrete application

1. General construction (residential, commercial, industrial)
2. Paving
3. High performance concrete
4. Precast concrete
5. Masonry and masonry units
6. Mass concrete
7. Shotcrete

Mixture proportion

The optimum mixture proportions for ternary blends, as with other concrete, will be dependent on the final use of the concrete, construction requirements and seasonal considerations. As with other concrete, cold weather will affect the early strength gain and mixture proportions may need to be adjusted to assure job-site performance. In low W/CM applications such as paving, mixtures with 15percent fly ash and 30% slag cement component have been used successfully.



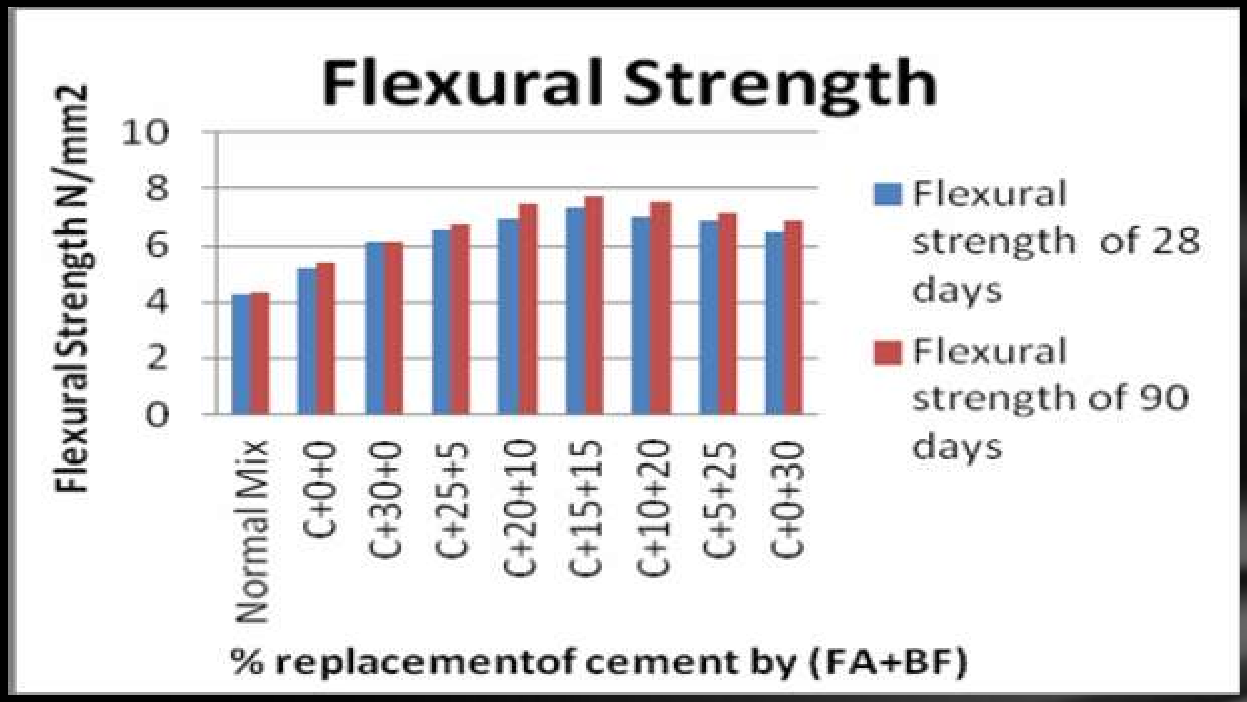
Fly ash, a powder resembling cement, has been used in concrete since the 1930s.

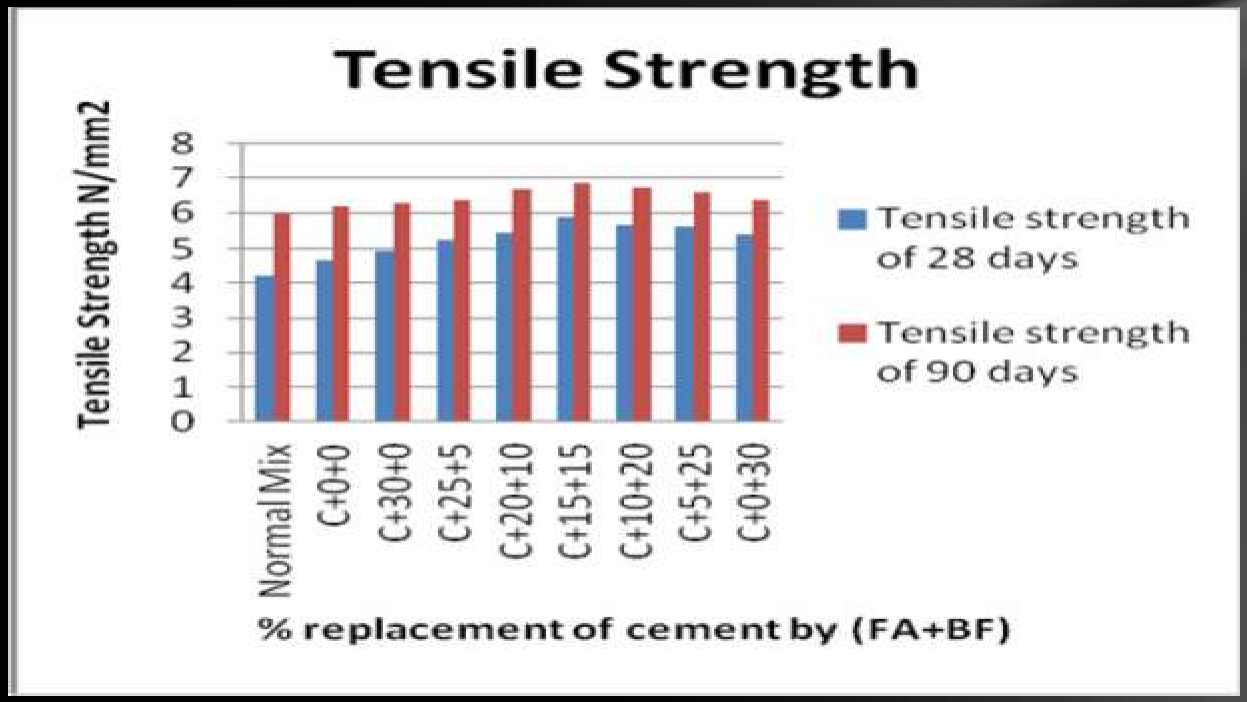


Ground granulated blast furnace slag



Silica fume powder


Flexural Strength
Flexural Strength N/mm2
10
8
6
4
2
0
Normal Mix
C+0+0
C+30+0
C+25+5
C+20+10
C+15+15
C+10+20
C+5+25
C+0+30
Flexural strength of 28 days
Flexural strength of 90 days
% replacementof cement by (FA+BF)


Tensile Strength
Tensile Strength N/mm2
8
7
6
5
4
3
2
1
0
Normal Mix
C+0+0
C+30+0
C+25+5
C+20+10
C+15+15
C+10+20
C+5+25
C+0+30
Tensile strength of 28 days
Tensile strength of 90 days
% replacement of cement by (FA+BF)

5.Light weight concrete

- Structural lightweight concrete is similar to normal weight concrete except that it has a lower density.
- Made with lightweight aggregates.
- Air-dry density in the range of 1350 to 1850 kg/m3
- 28-day compressive strength in excess of 17 Mpa.
- Structural lightweight concrete is used primarily to reduce the dead-load weight in concrete members, such as floors in high-rise buildings.
- **Structural Lightweight Aggregates:**

➢ Rotary kiln expanded clays, shales, and slates

➢ Sintering grate expanded shales and slates

➢ Pelletized or extruded fly ash

➢ Expanded slags

- **Compressive Strength:**

The compressive strength of structural lightweight concrete is usually **related to the cement content at a given slump and air content, rather than to a water-to-cement ratio.** This is due to the difficulty in determining how much of the total mix water is absorbed into the aggregate and thus not available for reaction with the cement.

- Slump:

1. Due to lower aggregate density, structural lightweight concrete does not slump as much as normal-weight concrete with the same workability.
2. A lightweight air-entrained mixture with a slump of 50 to 75 mm (2 to 3 in.) can be placed under conditions that would require a slump of 75 to 125 mm (3 to 5 in.)
3. With higher slumps, the large aggregate particles tend to float to the surface, making finishing difficult.

Entrained Air in light-weight concrete

1. As with normal-weight concrete, entrained air in structural lightweight concrete **ensures resistance to freezing and thawing and to deicer applications.**

2. It also **improves workability, reduces bleeding and segregation,** and may compensate for minor grading deficiencies in the aggregate.

3. Air contents are **generally between 5% and 8%, depending on the maximum size of coarse aggregate** (paste content) used and the exposure conditions.

Placing, Finishing, and Curing

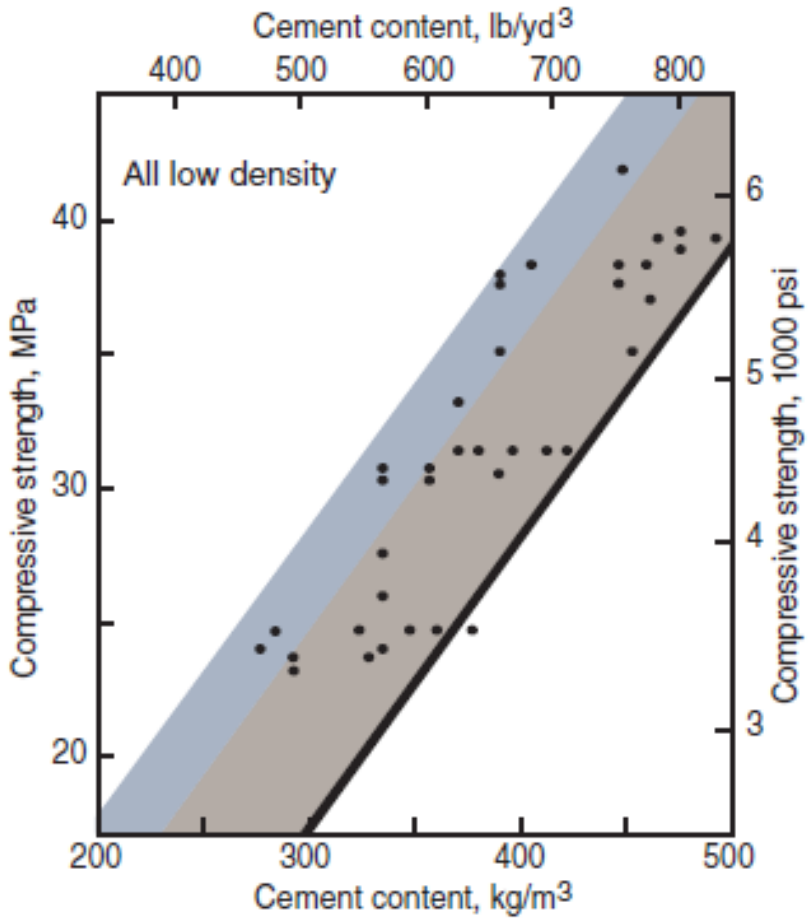
1. Structural lightweight concrete is generally easier to handle and place than normal-weight concrete.
2. A slump of 50 to 100 mm (2 to 4 in.) produces the best results for finishing.
3. If pumped concrete is being considered, the specified suppliers and contractor should all be consulted about performing a field trial using the pump and mixture planned for the project.
4. Adjustments to the mixture maybe necessary.
5. **pumping pressure causes the aggregate to absorb more water, thus reducing the slump and increasing the density of the concrete.**

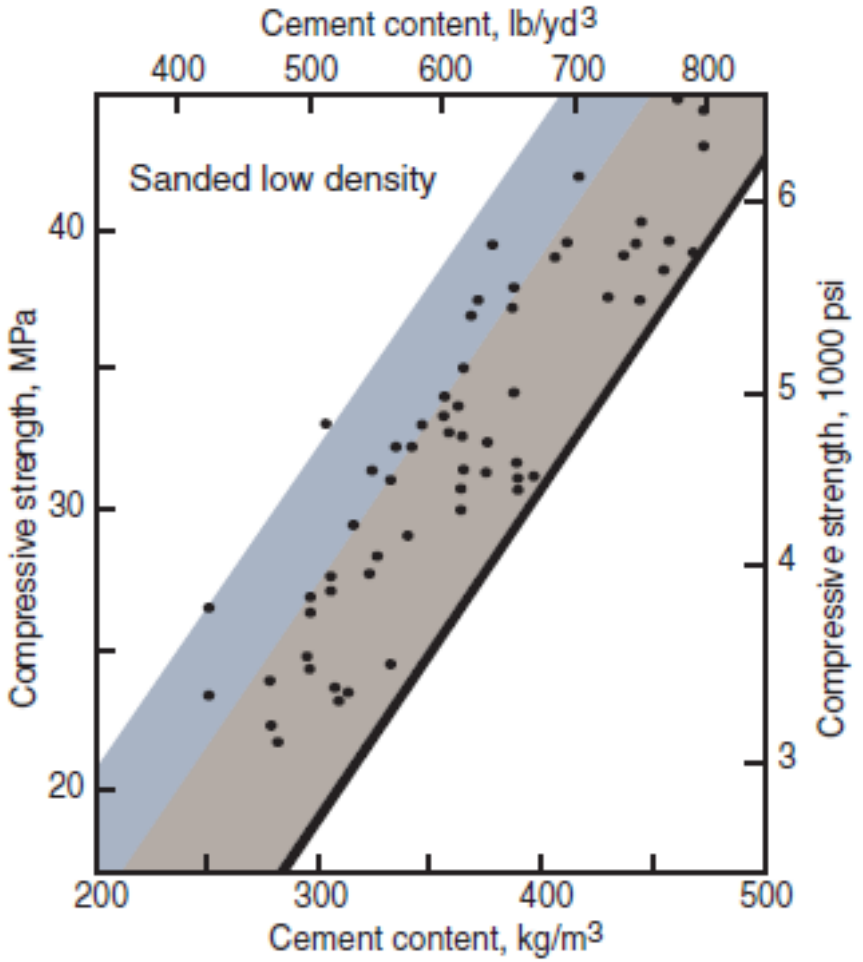




Placing and finishing of concrete.

Structural Lightweight Aggregate Concrete





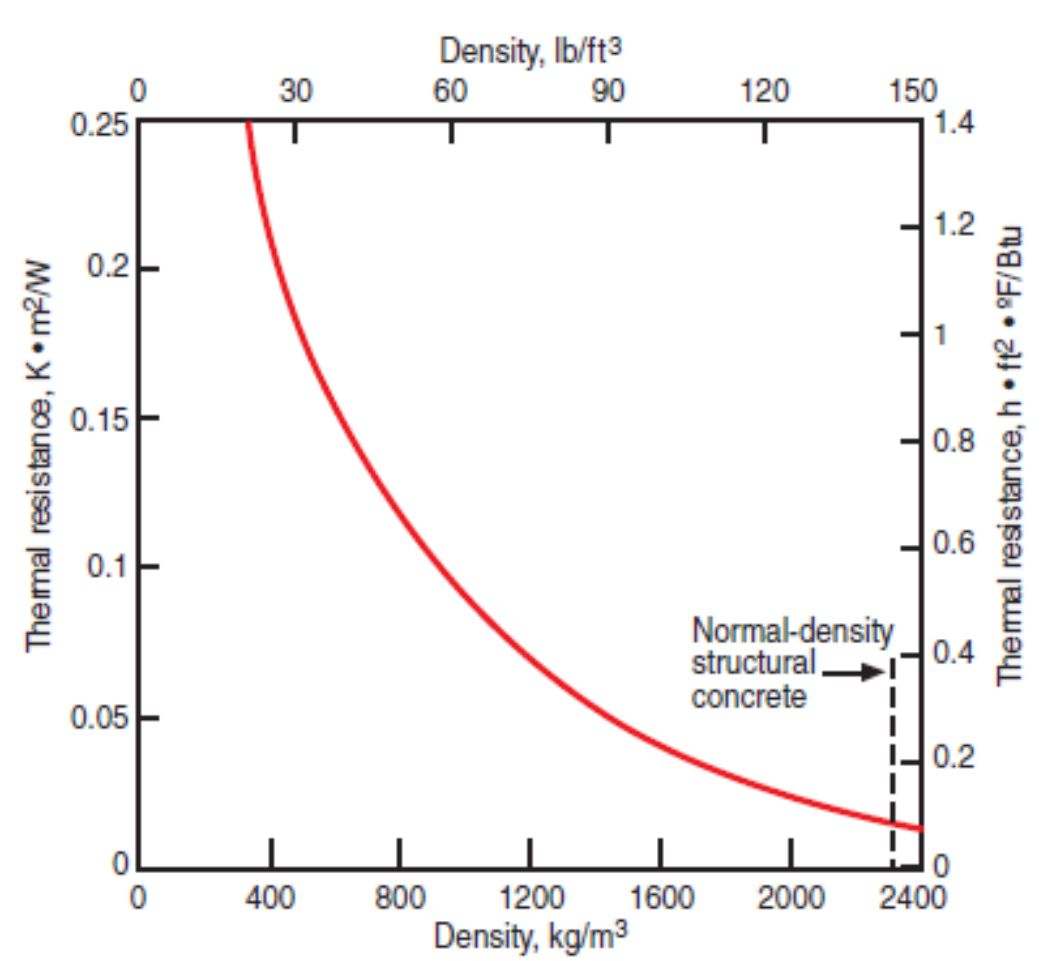
Structural Lightweight Aggregate Concrete

- Mixing
  - Similar to normal-weight concrete
  - Aggregates may need prewetting
- Workability
  - Lightweight concrete exhibits better workability for a given slump
- Finishing
  - Generally starts earlier
- Curing
  - Similar curing – added benefit of internal curing

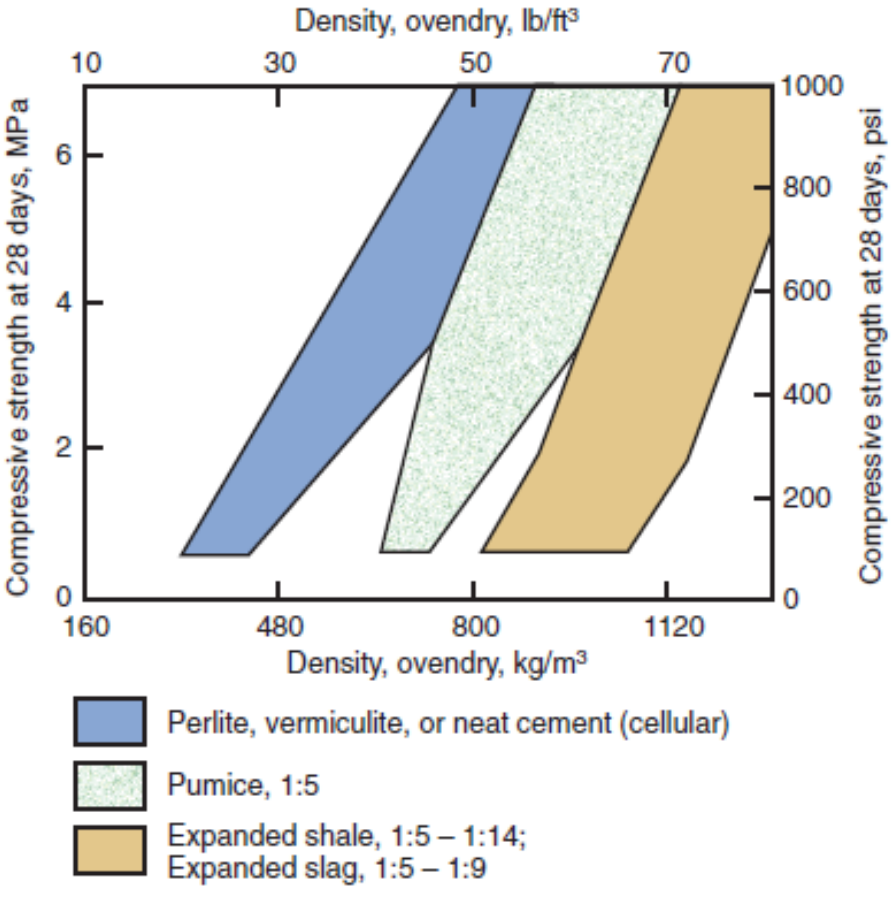
Type of concrete	Ratio: portland cement to aggregate by volume	Ovendry density, kg/m³ (pcf)	Type I portland cement, kg/m³ (lb/yd³)	Water-cement ratio, by mass	28-day compressive strength, MPa (psi), 150 x 300-mm (6 x 12-in.) cylinders
Perlite*	1:4	480 to 608 (30 to 38)	362 (610)	0.94	2.75 (400)
	1:5	416 to 576 (26 to 36)	306 (516)	1.12	2.24 (325)
	1:6	352 to 545 (22 to 34)	245 (414)	1.24	1.52 (220)
	1:8	320 to 512 (20 to 32)	234 (395)	1.72	1.38 (200)
Vermiculite*	1:4	496 to 593 (31 to 37)	380 (640)	0.98	2.07 (300)
	1:5	448 to 496 (28 to 31)	295 (498)	1.30	1.17 (170)
	1:6	368 to 464 (23 to 29)	245 (414)	1.60	0.90 (130)
	1:8	320 to 336 (20 to 21)	178 (300)	2.08	0.55 (80)
Polystyrene:** sand:					
0 kg/m³ (0 lb/yd³)	1:3.4	545 (34)‡	445 (750)	0.40	2.24 (325)
73 kg/m³ (124 lb/yd³)	1:3.1	625 (39)‡	445 (750)	0.40	2.76 (400)
154 kg/m³ (261 lb/yd³)	1:2.9	725 (44)‡	445 (750)	0.40	3.28 (475)
200 kg/m³ (338 lb/yd³)	1:2.5	769 (48)‡	474 (800)	0.40	3.79 (550)
Cellular* (neat cement)	—	625 (39)	524 (884)	0.57	2.41 (350)
	—	545 (34)	468 (790)	0.56	1.45 (210)
	—	448 (28)	396 (668)	0.57	0.90 (130)
	—	368 (23)	317 (535)	0.65	0.34 (50)
Cellular† (sanded)††	1:1	929 (58)	429 (724)	0.40	3.17 (460)
	1:2	1250 (78)	373 (630)	0.41	5.66 (820)
	1:3	1602 (100)	360 (602)	0.51	15.10 (2190)

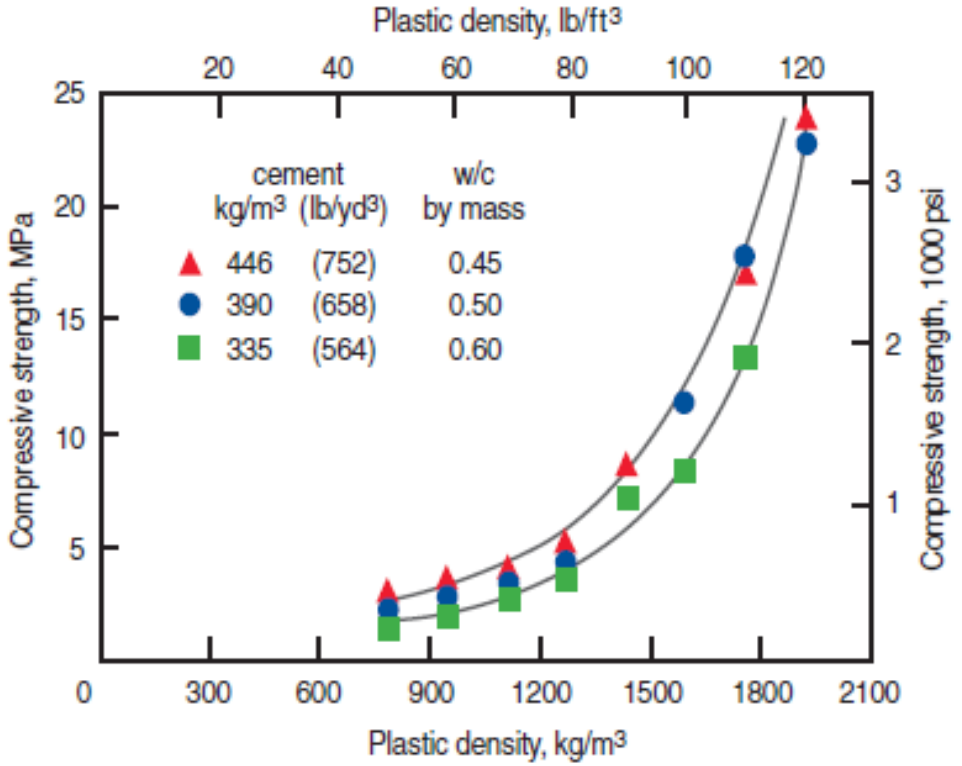
* Reichard (1971).
** Source: Hanna (1978). The mix also included air entrainment and a water-reducing agent.
† Source: Gustaferro and others (1970).
†† Dry-rodded sand with a bulk density of 1600 kg/m³ (100 pcf).
‡ Air-dry density at 28 days, 50% relative humidity.

Insulating and Moderate-Strength Lightweight Concretes



Insulating and Moderate-Strength Lightweight Concretes





6.Polymer concrete

- Polymer concrete is part of group of concretes that use polymers to supplement or replace cement as a binder.
- The types include polymer-impregnated concrete, polymer concrete, and polymer-Portland- cement concrete.
- In polymer concrete, thermosetting resins are used as the principal polymer component due to their high thermal stability and resistance to a wide variety of chemicals.
- Polymer concrete is also composed of aggregates that include silica, quartz, granite, limestone, and other high quality material.

- Polymer concrete may be **used for new construction or repairing of old concrete.**
- The low permeability and corrosive resistance of polymer concrete allows it to be used in swimming pools, sewer structure applications, drainage channels, electrolytic cells for base metal recovery
- It is especially **suited to the construction and rehabilitation of manholes due to their ability to withstand toxic and corrosive sewer gases and bacteria** commonly found in sewer systems.
- It can also be used as a replacement for asphalt pavement, for higher durability and higher strength.
- Polymer concrete has historically not been widely adopted due to the high costs and difficulty associated with traditional manufacturing techniques.

Advantages & Disadvantages

Advantages

1. Rapid curing at ambient temperatures
2. High tensile, flexural, and compressive strengths
3. Good adhesion to most surfaces
4. Good long-term durability with respect to freeze and thaw cycles
5. Low permeability to water and aggressive solutions
6. Good chemical resistance
7. Good resistance against corrosion
8. Lighter weight (only somewhat less dense than traditional concrete, depending on the resin content of the mix)
9. May be vibrated to fill voids in forms
10. Allows use of regular form-release agents (in some applications)
11. Dielectric

Disadvantages

1. Product hard to manipulate with conventional tools such as drills and presses due to its strength and density. Recommend getting pre-modified product from the manufacturer
2. Small boxes are more costly when compared to its precast counterpart however pre cast concretes induction of stacking or steel covers quickly bridge the gap.





Using of polymer concrete on old concrete

Photocatalytic Concrete

- Titanium-based additions act as catalysts in sunlight
  - Breaks down pollution into oxygen, water, CO_2, nitrate, sulfate
- Well-suited for structures with high sun exposure
- Can be used for streets
  - Study shows NO_x reduction of 25-45% over control pavement

7.Self compacting concrete

- Self-compacting concrete (SCC) is an innovative concrete that does not require vibration for placing and compaction.
- It is able to flow under its own weight, completely filling formwork and achieving full compaction, even in the presence of congested reinforcement.
- Very close to the Kolhapur there is project of steel industry, sand used for the formation of mould when the moulds are opened the waste sand is dumped for the filling the low lying areas while doing this the agriculture areas is converted into barren area.

- Because there is no space for the waste other than the land filling. similar case is in case of aluminium industry where red mud is concluded to be waste, which contains lot amount of bauxite and that is why red mud is also dump in the nearby areas here it is causing big threat for the society and it is disturbing the eco system of the environment.
- So it is the need to use this particular otherwise waste material for the constructive in such fashion in the case of concrete so that concrete which became cost effective as well as eco-friendly.

- **Types:**

1. **Powder type of self-compacting concrete**: This is proportioned to give the required self-compactability by reducing the water-powder ratio and provide adequate segregation resistance.
2. **Viscosity agent type self-compacting concrete**: This type is proportioned to provide self-compaction by the use of viscosity modifying admixture to provide segregation resistance.
3. **Combination type self-compacting concrete**: This type is proportioned so as to obtain self-compactability mainly by reducing the water powder ratio.

- **Fresh SCC Properties:**

1. Filling ability (excellent deformability)
2. Passing ability (ability to pass reinforcement without blocking)
3. High resistance to segregation.





Placing of self compacting concrete

8.Coloured concrete:

- **Coloured concrete can be produced by using coloured aggregates or by adding colour pigments (ASTM C 979) or both.**
- If surfaces are to be washed with acid, a delay of approximately two weeks after casting is necessary.
- Coloured aggregates may be natural rock such as quartz, marble, and granite, or they may be ceramic materials, synthetic pigments generally give more uniform results.
- The amount of colour pigments added to a concrete mixture should not be more than 10% of the mass of the cement.





Coloured concrete

Colored Concrete



9. **Fibre reinforced concrete**

- Fibre reinforced concrete (FRC) may be defined as a composite materials made with Portland cement, aggregate, and incorporating discrete discontinuous fibres.
- The role of randomly distributes discontinuous fibres is to bridge across the cracks that develop provides some post-cracking “ductility”.
- The real contribution of the fibres is to increase the toughness of the concrete under any type of loading.
- The fibre reinforcement may be used in the form of three – dimensionally randomly distributed fibres throughout the structural member when the added advantages of the fibre to shear resistance and crack control can be further utilised.

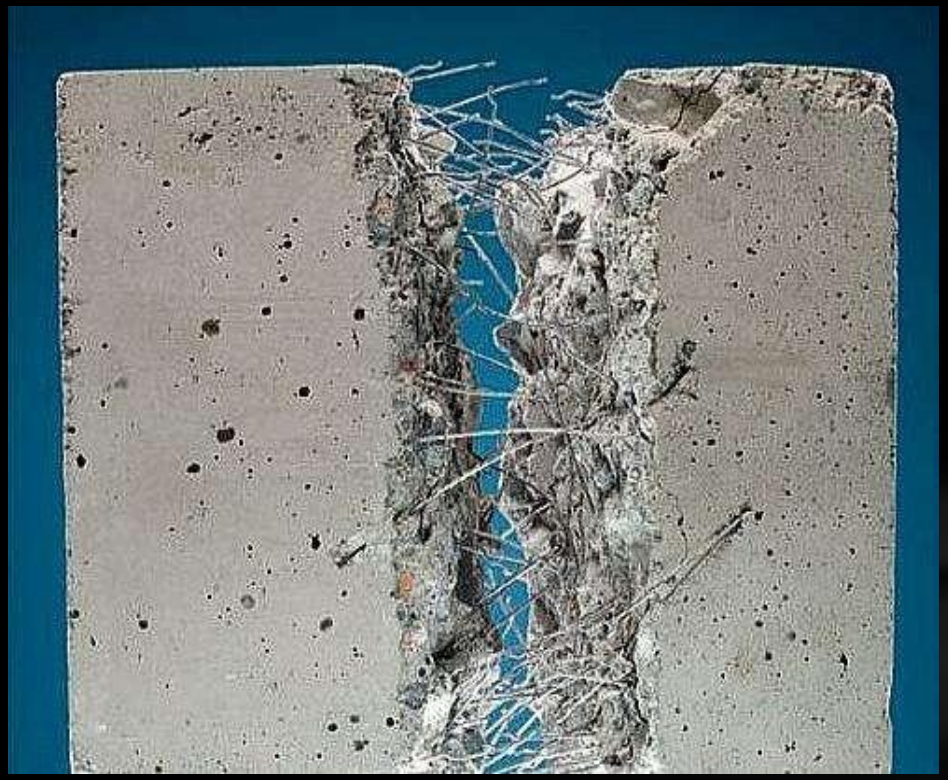
- **Tensile Strength:**

1. Fibres aligned in the direction of the tensile stress may bring about very large increases in direct tensile strength, as high as 1.33% for 5% of smooth, straight steel fibres.
2. Thus, adding fibres merely to increase the direct tensile strength is probably worthwhile.
3. However, as in compression, steel fibres do lead to major increases in the post cracking behaviour or toughness of the composites.

- **Application of SFRC:**

- The most common applications are

1. pavements
2. tunnel linings
3. pavements and slabs
4. shotcrete
5. shotcrete also containing silica fume, airport pavements, bridge deck slab repairs





Fibre reinforced concrete

10.Pervious concrete:

- Pervious (porous or no-fines) concrete contains a narrowly graded coarse aggregate, little to no fine aggregate, and insufficient cement paste to fill voids in the coarse aggregate.
- Low water-cement ratio, low-slump concrete resembling popcorn held together by cement paste.
- Produces a concrete with a high volume of voids (20% to 35%) and a high permeability that allows water to flow through it easily below.





Pervious concrete

- Pervious concrete is used in hydraulic structures as drainage media, and in parking lots, pavements, and airport local groundwater supply by allowing water to penetrate the concrete to the ground
- Pervious concretes have also been used in tennis courts and greenhouses.
- The compressive strength of different mixes can range from 3.5 to 27.5 Mpa.
- Drainage rates commonly range from 100 to 900 lit.per minute per square meter.

11. Water-proof concrete:

- Top Proof waterproof concrete contains two specially formulated admixtures.
- The first reduces the water/cement ratio, increasing the density of the mix and minimising the size of the pores.
- The second fills the remaining pores ensuring a completely watertight finish. This means there is no need for external membranes, reducing cost and labour.

Waterproof concrete benefits:

1. No need for external membranes, reducing cost and labour
2. Water resistant, quality-assured waterproof concrete
3. Less susceptible to cracking
4. No minimum order size
5. 100% recyclable

Waterproof concrete applications:

1. Basements
2. Underground car parks
3. Electrical and other plant rooms
4. Swimming pools
5. Aquariums and aquatic centres
6. Waterside buildings



Wave cutters made up of water-proof concrete

12. **Mass concrete:**

- "**An large volume of cast-in-place concrete with dimensions large enough to require that measures be taken to cope with the generation of heat and attendant volume change to minimize cracking.**"
- Mass concrete includes not only low-cement-content concrete used in dams and other massive structures but also moderate to high cement content concrete in structural members of bridges and buildings.
- **As the interior concrete increases in temperature and expands, the surface concrete may be cooling and contracting.**
- The width and depth of cracks depends upon the temperature differential, physical properties of the concrete, and the reinforcing steel.



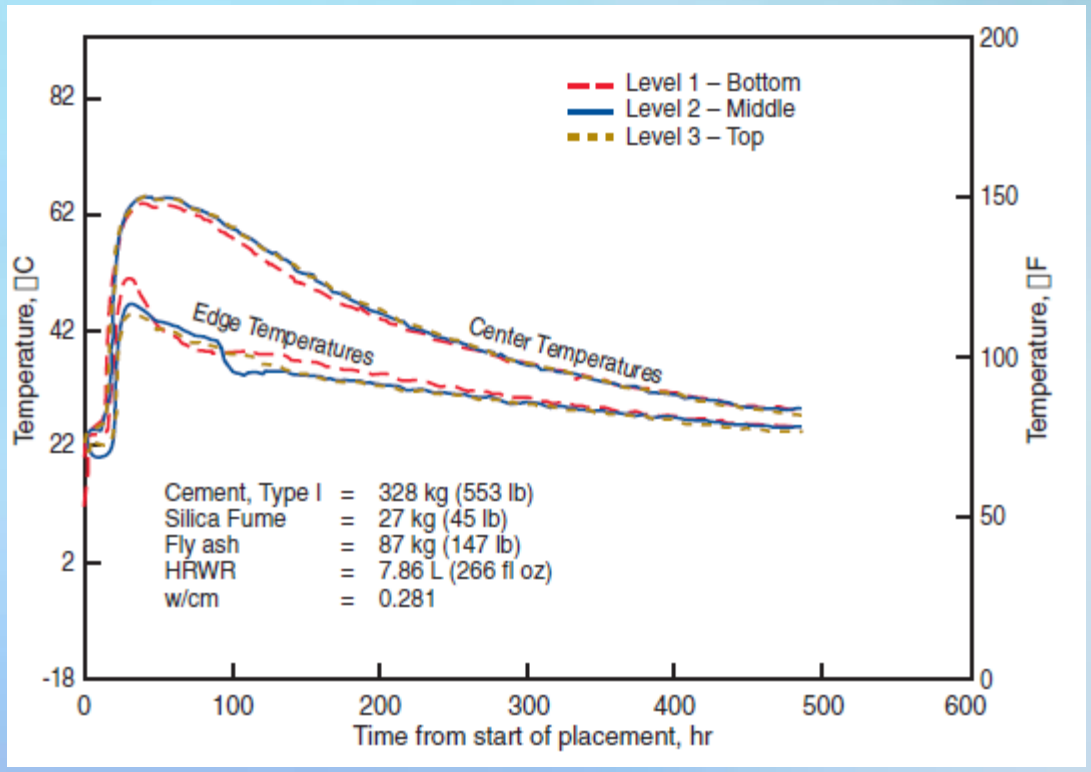
Casting of Mass concrete

Mass Concrete

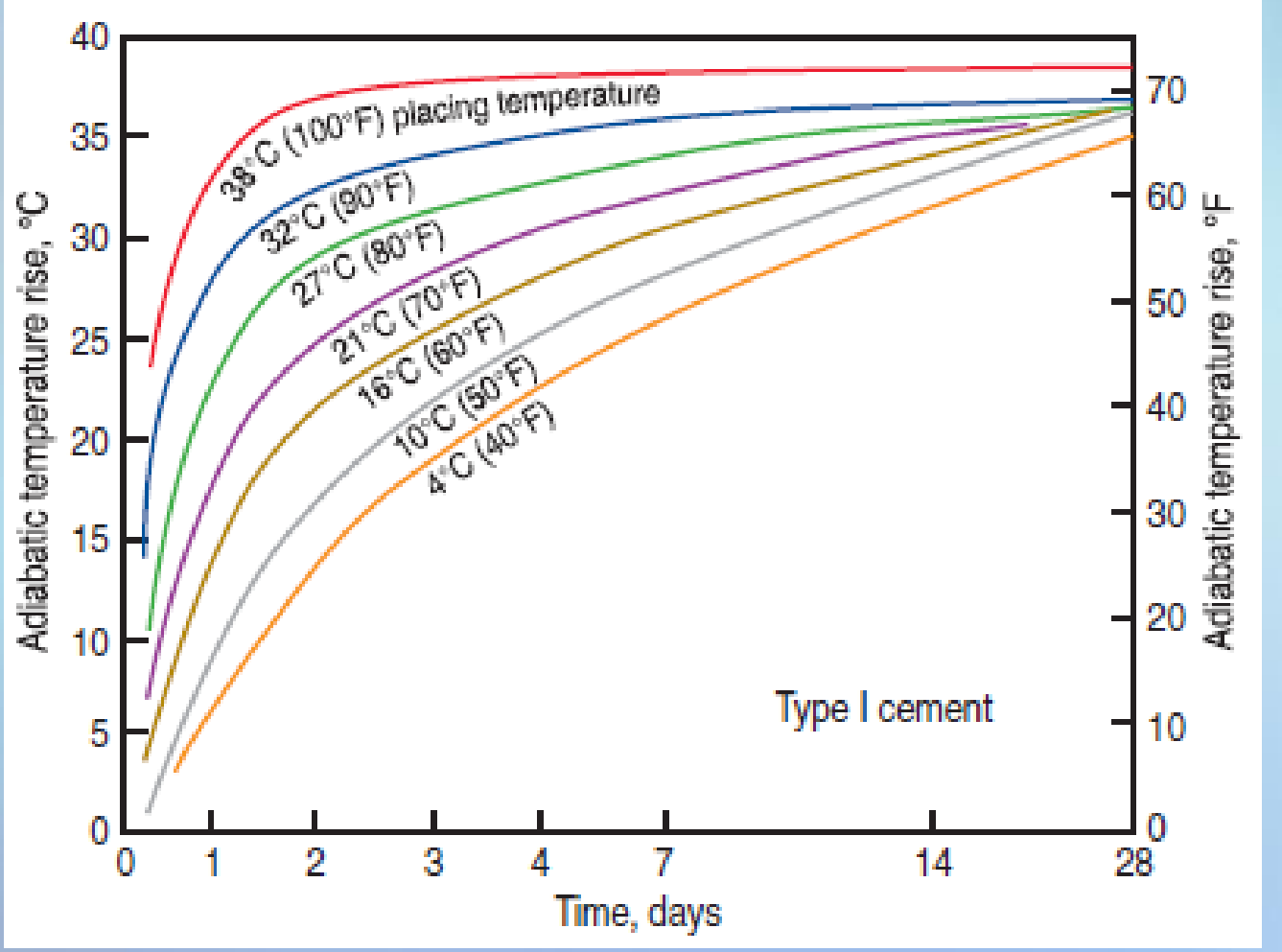


Mass Concrete

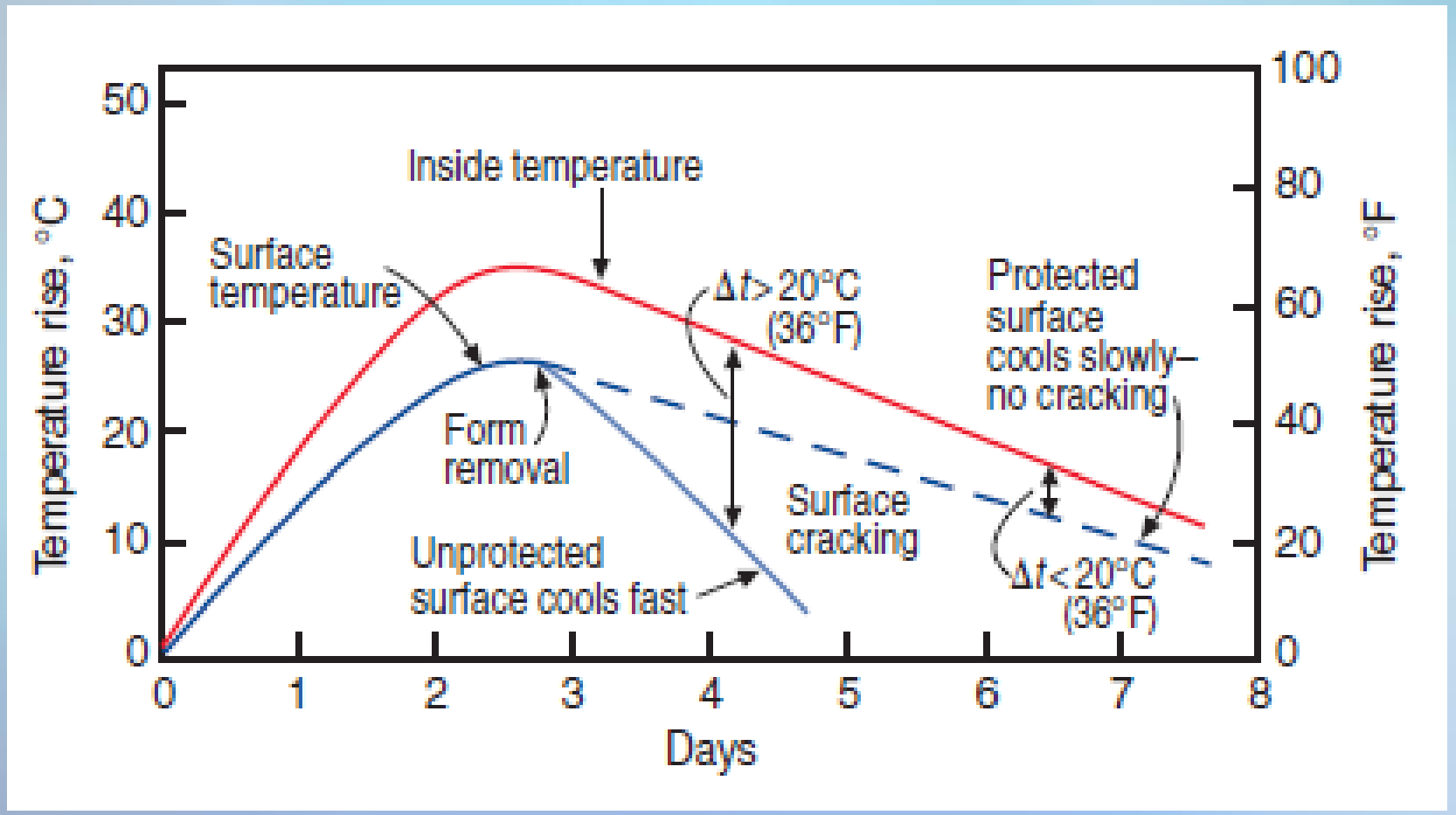




Mass Concrete



Mass Concrete



PCA
Portland Cement Association

Autoclaved Cellular Concrete





High-Density Concrete

Type of aggregate	Fixed-water,* percent by weight	Aggregate relative density	Aggregate bulk density, kg/m³ (pcf)	Concrete density, kg/m³ (pcf)
Goethite	10–11	3.4–3.7	2080–2240 (130–140)	2880–3200 (180–200)
Limonite**	8–9	3.4–4.0	2080–2400 (130–150)	2880–3360 (180–210)
Barite	0	4.0–4.6	2320–2560 (145–160)	3360–3680 (210–230)
Ilmenite	†	4.3–4.8	2560–2700 (160–170)	3520–3850 (220–240)
Hematite	†	4.9–5.3	2880–3200 (180–200)	3850–4170 (240–260)
Magnetite	†	4.2–5.2	2400–3040 (150–190)	3360–4170 (210–260)
Ferrophosphorus	0	5.8–6.8	3200–4160 (200–260)	4080–5290 (255–330)
Steel punchings or shot	0	6.2–7.8	3860–4650 (230–290)	4650–6090 (290–380)

* Water retained or chemically bound in aggregates.
** Test data not available.
† Aggregates may be combined with limonite to produce fixed-water contents varying from about ½% to 5%.

No-Slump Concrete

- Defined by slump less than 6 mm (¼ in.)
- Slump cone is impractical
  - Vebe apparatus
  - Compacting-factor test
  - Thaulow drop table
- Air entrainment is still used for freeze-thaw resistance

Roller-Compacted Concrete



Soil-Cement



PCA
Portland Cement Association

White Concrete



Questions

